Sedov–Taylor Expansion from a Center of Expansion Beyond the Horizon: Enclosed-Mass Gravity and the Angular Pattern of the X-ray Cluster L_X–T Anisotropy

Chris

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Abstract

We consider a cosmological model in which the observed expansion is the interior flow of a Sedov–Taylor self-similar blast wave launched from a real Center of Expansion (CoE) that lies beyond the observable horizon. The observer is displaced from the CoE, the matter density is centrally concentrated and decreases outward, and gravity is computed from the enclosed mass via Gauss's law, $g = -GM(< r)/r^2$. Purely homologous expansion yields an isotropic apparent Hubble law for every observer, independent of position; direction dependence therefore arises only through the gravitational deceleration of the flow in the radially decreasing density profile. At leading order in the ratio of source depth to CoE distance the induced sky pattern is a quadrupole about the CoE axis, with a natural amplitude set by $\kappa = \Omega_m/2$. We compare the predicted angular pattern with the longitude dependence of the normalization A of the X-ray cluster L_X–T relation reported by Migkas et al. (2020). An empirical dipole-plus-quadrupole description of the digitized $A(\ell)$ profile attains $R^2 = 0.814$, compared with $R^2 = 0.271$ for a dipole alone. The axisymmetric Sedov–Taylor/Gauss geometry attains $R^2 = 0.307$; with a phase-rotation template and a multiplicative second-harmonic correction the agreement reaches $R^2 = 0.875$. The four extrema of the corrected profile, mapped onto the sky as small circles about a CoE axis tilted by 15° from the Galactic plane, pass close to the published extrema of the Migkas et al. A map. The phase-rotation and second-harmonic terms are presently empirical templates; deriving them from time-integrated gravity is the principal task for future work.

1 Introduction

The self-similar blast wave produced by a point release of energy E into a uniform medium of density ρ was obtained independently by Sedov [1] and Taylor [2]. Its shock radius obeys $R \propto (Et^2/\rho)^{1/5}$, and the solution generalizes to power-law ambient densities and, in the ultra-relativistic regime, to the similarity solutions of Blandford and McKee [3], which constitute the relativistic generalization of the Sedov–Taylor problem. In a cosmological setting, Bertschinger [4] found self-similar blast waves in an Einstein–de Sitter background whose early evolution follows the Sedov solution before being modified by the cosmological expansion. In general relativity, Smoller and Temple [5, 6] constructed exact solutions of the Einstein equations in which a Friedmann–Robertson–Walker expansion emerges behind a shock wave at the leading edge of a large-scale explosion, with the shock located at or beyond one Hubble length and, in the latter case, inside a black hole.

These works motivate a class of models in which the observed Hubble expansion is the interior flow of an explosion about a physical center, rather than a homogeneous expansion without a center.

In such a model the observer is, in general, not located at the center, and the question arises whether the resulting departures from isotropy are observable.

A candidate observable is the anisotropy of galaxy-cluster scaling relations reported by Migkas et al. [7]. Using 313 clusters from the Meta-Catalogue of X-ray detected Clusters of galaxies, these authors found that the normalization A of the L_X–T relation depends on sky direction, with anisotropy at the $\gtrsim 4\sigma$ level toward Galactic coordinates $(\ell, b) \sim (280^\circ, -20^\circ)$. Since L_X depends on the assumed luminosity distance while the spectroscopic temperature does not, a direction-dependent A can be interpreted as a direction-dependent distance–redshift relation.

In this paper we formulate a Sedov–Taylor model with an off-center observer, a centrally concentrated density profile, and enclosed-mass gravity (Section 2); describe the comparison with the Migkas et al. longitude profile $A(\ell)$ and sky map (Section 3); present the resulting fits and sky geometry (Section 4); and discuss the status and limitations of the comparison (Section 5).

2 Model

2.1 Dynamics: Sedov–Taylor self-similar expansion

We take the expansion to be the interior of a Sedov–Taylor blast wave centered on the CoE. For energy E released into a uniform medium of density ρ, dimensional analysis and self-similarity give the shock radius

$$R(t) = \xi_0 \left(\frac{E\, t^2}{\rho} \right)^{1/5}, \tag{2.1}$$

where ξ_0 is a dimensionless constant of order unity that depends on the adiabatic index [1, 2]. For a centrally concentrated medium whose density decreases as a power law,

$$\rho(r) = \rho_1 \left(\frac{r}{r_1} \right)^{-w}, \tag{2.2}$$

the same argument, with $\rho_1 r_1^w$ replacing ρ as the dimensional parameter, gives

$$R(t) = \xi_w \left(\frac{E\, t^2}{\rho_1 r_1^w} \right)^{1/(5-w)} \propto\ t^{2/(5-w)}, \qquad w < 5, \tag{2.3}$$

which reduces to Eq. (2.1) for $w = 0$. The shock velocity is $\dot{R} = [2/(5-w)]\, R/t$, so that the expansion rate at the shock is

$$H(t) \equiv \frac{\dot{R}}{R} = \frac{2}{(5-w)\, t}. \tag{2.4}$$

The corresponding relativistic self-similar solutions, including ambient densities varying with radius, were given by Blandford and McKee [3]; cosmological and general-relativistic analogues are those of Bertschinger [4] and of Smoller and Temple [5, 6].

In the interior of the blast we adopt the homologous approximation, in which every fluid element moves radially away from the CoE with a velocity proportional to its distance,

$$\vec{v}(\vec{R}) = H\, \vec{R}. \tag{2.5}$$

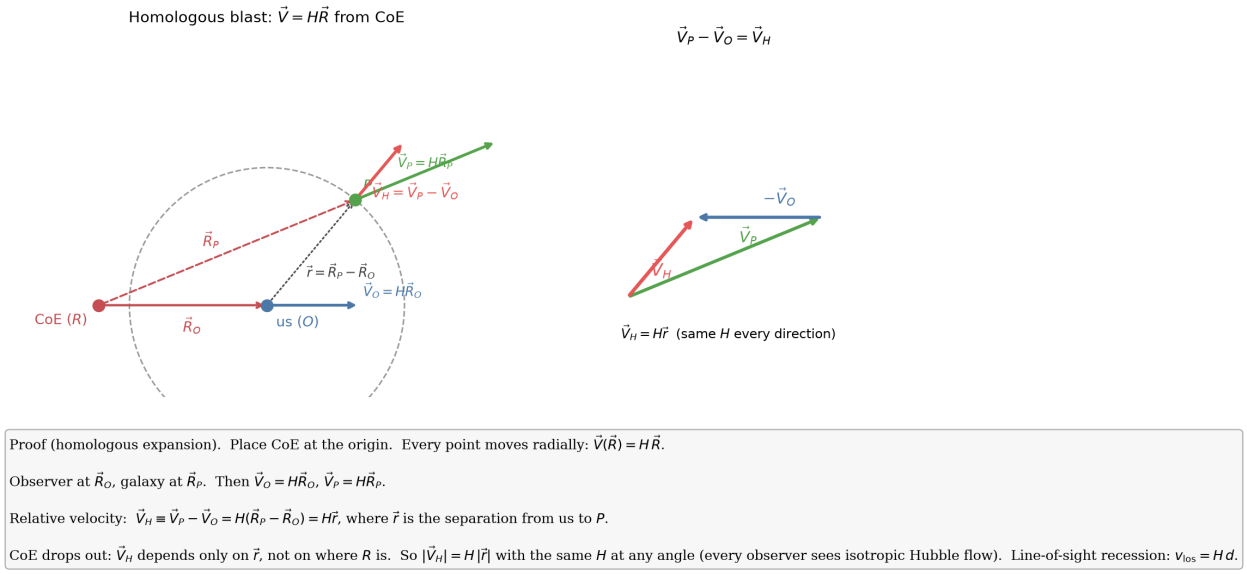
Let the observer O be located at $\vec{R}_O$ and a source P at $\vec{R}_P$. The relative velocity is

$$\vec{v}_{\rm rel} = \vec{v}_P - \vec{v}_O = H\, (\vec{R}_P - \vec{R}_O) = H\, \vec{d}, \tag{2.6}$$

where $\vec{d}$ is the observer–source separation. The line-of-sight recession velocity is $v_{\rm los} = \vec{v}_{\rm rel} \cdot \hat{d} = Hd$, so that

$$H_{\rm obs} \equiv \frac{v_{\rm los}}{d} = H \tag{2.7}$$

in every direction. The position of the CoE therefore drops out of all relative velocities (Fig. 1): an off-center observer in a purely homologous flow measures an isotropic Hubble law. Any observable anisotropy must arise from a departure from homology, which in the present model is supplied by gravity.



Proof (homologous expansion). Place CoE at the origin. Every point moves radially: $\vec{V}(\vec{R}) = H\vec{R}$.

Observer at $\vec{R}_O$, galaxy at $\vec{R}_P$. Then $\vec{V}_O = H\vec{R}_O$, $\vec{V}_P = H\vec{R}_P$.

Relative velocity: $\vec{V}_H \equiv \vec{V}_P - \vec{V}_O = H(\vec{R}_P - \vec{R}_O) = H\vec{r}$, where $\vec{r}$ is the separation from us to P.

CoE drops out: $\vec{V}_H$ depends only on $\vec{r}$, not on where R is. So $|\vec{V}_H| = H\,|\vec{r}|$ with the same H at any angle (every observer sees isotropic Hubble flow). Line-of-sight recession: $v_{\rm los} = H\,d$.

Figure 1: Homologous expansion about the CoE. Absolute velocities $\vec{V} = H\vec{R}$ are directed radially away from the CoE, but the velocity of a source P relative to the observer O is $H(\vec{R}_P - \vec{R}_O)$, which depends only on the separation. Every observer therefore measures the same isotropic expansion rate H, irrespective of its distance from the CoE (Eqs. 2.5–2.7).

2.2 Geometry

The CoE is placed at the origin and lies beyond the observer's horizon. The observer is at distance D from the CoE along the unit vector $\hat{D}$, and sources are located at observer-centric distance d in direction $\hat{n}$, so that the source position relative to the CoE is $\vec{r} = \vec{D} + d\,\hat{n}$ (Fig. 2). We write

$$\mu \equiv \hat{n} \cdot \hat{D} = \cos\phi, \tag{2.8}$$

where ϕ is the sky angle measured from the anti-CoE direction ($\phi = 0$) toward the CoE ($\phi = 180°$). The relevant expansion parameter is d/D, which is small for survey depths much less than the distance to the CoE.

2.3 Density profile

A uniform-density sphere produces an interior gravitational acceleration $\vec{g} \propto -\vec{r}$, which is itself homologous; it decelerates the flow uniformly and cannot generate direction dependence in $H_{\rm obs}$.

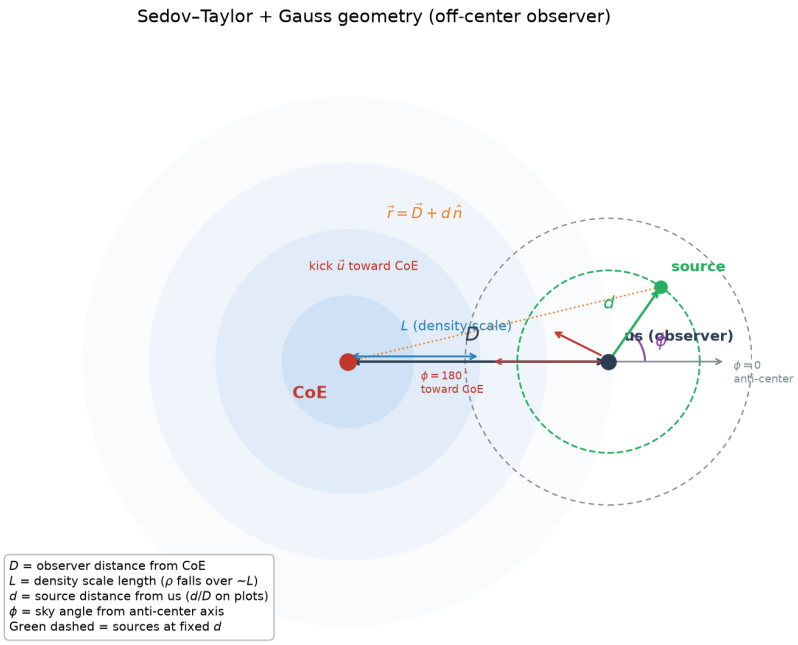


Figure 2: Model geometry. The CoE lies at the origin, embedded in a centrally concentrated density distribution with scale length L. The observer is at distance D from the CoE; sources lie at observer-centric distance d and sky angle ϕ measured from the anti-CoE direction.

We therefore adopt a spherically symmetric, centrally concentrated profile that decreases outward. For the calculations presented here we use the exponential profile

$$\frac{\rho(r)}{\rho_0} = e^{-r/L}, \tag{2.9}$$

whose local logarithmic slope $w_{\rm eff} = -\,d\ln\rho/d\ln r = r/L$ increases outward, connecting to the power-law scaling of Eq. (2.3). The enclosed-mass fraction, with $X \equiv r/L$ and M the total mass, is

$$\frac{M(<r)}{M} = 1 - e^{-X}\left(1 + X + \tfrac{1}{2}X^2\right), \tag{2.10}$$

obtained by integrating $4\pi r^2\rho(r)$. Both functions are shown in Fig. 3.

2.4 Gravity

For a spherically symmetric mass distribution, Gauss's law gives the gravitational acceleration at radius r from the CoE in terms of the enclosed mass alone,

$$\vec{g}(r) = -\frac{G\,M(<r)}{r^2}\,\hat{r}. \tag{2.11}$$

Because $M(<r)/r^3$ decreases outward for the profile of Eq. (2.9), sources closer to and farther from the CoE than the observer are decelerated by different amounts relative to the observer, which breaks the homology of Eq. (2.5). A natural dimensionless measure of the gravitational strength at the observer is

$$\kappa \equiv \frac{G\,M(<D)}{D^3H^2} = \frac{4\pi G\bar{\rho}(<D)}{3H^2} = \frac{\Omega_m}{2}, \tag{2.12}$$

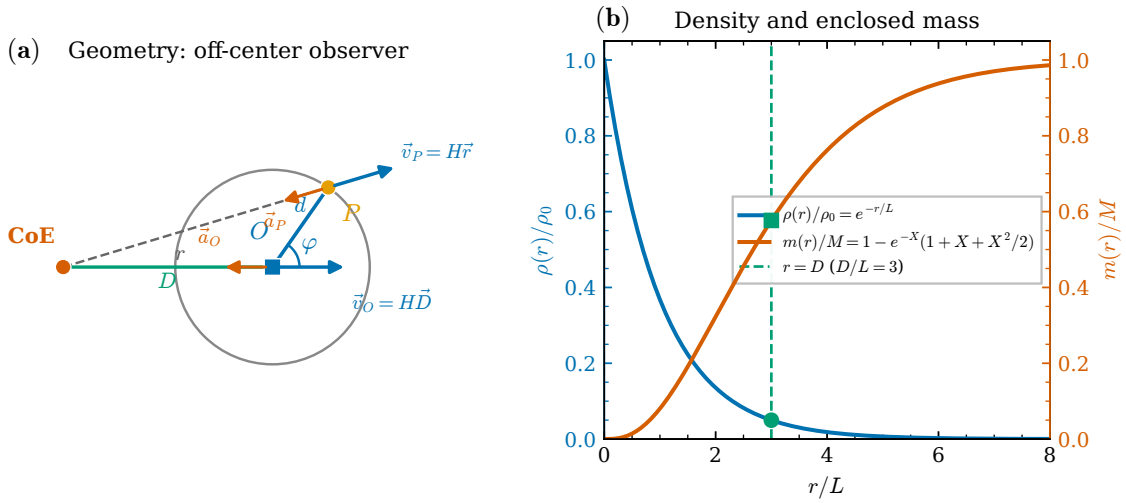


Figure 3: Centrally concentrated density profile used in the model. Left: exponential density $\rho(r)/\rho_0 = e^{-r/L}$ (Eq. 2.9). Right: enclosed-mass fraction $M(< r)/M$ (Eq. 2.10). The vertical marker indicates an illustrative observer distance $D/L = 3$.

where $\bar{\rho}(< D)$ is the mean density within D and $\Omega_m = 8\pi G\bar{\rho}/(3H^2)$. For $\Omega_m = 0.3$ this gives $\kappa_{\rm nat} = 0.15$.

Expanding the relative gravitational acceleration between source and observer to leading orders in d/D gives the tidal form

$$\frac{H_{\rm obs}(\hat{n})}{H} - 1 \simeq \alpha\kappa\,(3\mu^2 - 1) + \alpha\kappa\,\frac{d}{D}\left(\tfrac{9}{2}\mu - \tfrac{15}{2}\mu^3\right), \tag{2.13}$$

where α is an order-unity coefficient that depends on how the acceleration is integrated along the flow. The leading term is a quadrupole, $3\mu^2 - 1 = 2P_2(\mu)$, with Legendre coefficient $C_2 \simeq 2\alpha\kappa$ ($C_2 \simeq 0.30$ for $\alpha = 1$, $\kappa = 0.15$), while the full-sky dipole coefficient C_1 vanishes at this order; the first odd contribution is an octupole of relative size $\mathcal{O}(d/D)$. A spherically symmetric enclosed-mass model with a distant CoE therefore predicts a sky pattern that is axisymmetric about $\hat{D}$ and quadrupole-dominated.

3 Methods: comparison with Migkas et al. (2020)

3.1 Observational target

Migkas et al. [7] report the L_X–T normalization A, relative to its all-sky value $A_{\rm all}$, as a function of Galactic longitude in sliding longitude sectors (their Fig. 6), and as a full-sky map of $A/A_{\rm all}$ from cones of opening angle $\theta = 75°$ (their Fig. 8). We digitized the longitude profile at 71 longitudes spaced by 5° ($\ell = 5°, \ldots, 355°$). For the sky comparison we use the extrema of the $\theta = 75°$ map: the highest normalization near $(\ell, b) = (17°, -9°)$ and the lowest near $(274°, -22°)$ [7].

As an empirical characterization of the digitized profile, a linear least-squares fit of a monopole plus dipole in longitude gives $R^2 = 0.271$, whereas adding the second harmonic,

$$\frac{A(\ell)}{A_{\rm all}} = 0.997 + 0.030\cos\ell + 0.043\sin\ell + 0.064\cos 2\ell + 0.037\sin 2\ell, \tag{3.1}$$

gives $R^2 = 0.814$ and an rms residual of 0.031 (Fig. 4). The second harmonic is thus the dominant angular structure of the profile, consistent in character with the quadrupole predicted by Eq. (2.13).

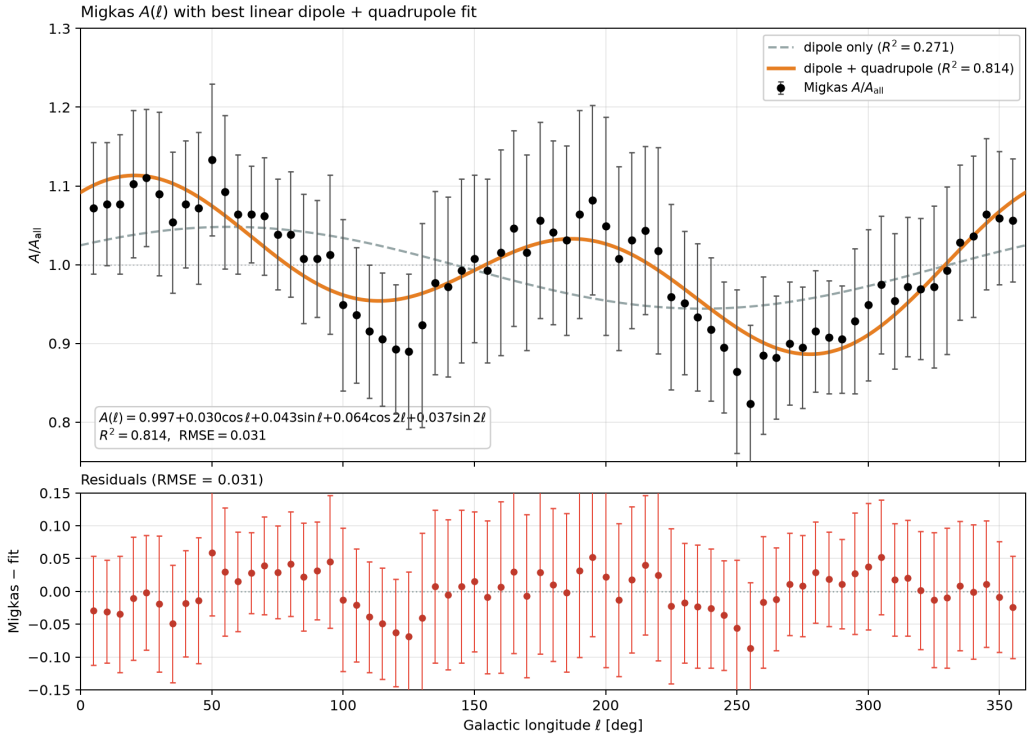


Figure 4: Digitized longitude profile $A/A_{\rm all}$ of Migkas et al. [7] (points with error bars) with linear least-squares fits. Dashed: monopole plus dipole, $R^2 = 0.271$. Solid: monopole plus dipole plus quadrupole (Eq. 3.1), $R^2 = 0.814$, rms residual 0.031. Lower panel: residuals of the dipole-plus-quadrupole fit.

3.2 Model observable

The model predicts $H_{\rm obs}(\hat{n})/H$. We map this onto the normalization through

$$\frac{A}{A_{\rm all}} \propto \frac{H_{\rm obs}}{H}, \tag{3.2}$$

renormalized to unit mean over the longitude strip. The model profile is evaluated along the Galactic plane, with the angle ϕ of Eq. (2.8) related to Galactic longitude by the orientation of the CoE axis.

3.3 Phase-rotation and second-harmonic templates

Two empirical corrections are applied to the axisymmetric model profile $A_{\rm phys}(\ell)$. The first is a phase-rotation template,

$$A'(\ell) = \left[1 - \mathrm{Amp}\,\sin(\Omega\ell + \varphi)\right] A_{\rm phys}(\ell), \qquad \Omega = \frac{2\pi}{360^\circ}, \tag{3.3}$$

with amplitude Amp and phase φ scanned on a grid. The second is a multiplicative second-harmonic correction,

$$A_{\rm c}(\ell) = \left[1 + \varepsilon\,\sin(2\ell + \psi)\right] A'(\ell), \tag{3.4}$$

with ε and ψ fitted to the residuals. Goodness of fit is reported as the coefficient of determination R^2 against the digitized profile, together with the rms residual.

3.4 Sky mapping of extrema

The extrema of the corrected profile $A_{\rm c}(\ell)$ define four angles θ, measured from the anti-CoE direction in the model frame. Because the model is axisymmetric about $\hat{D}$, each angle corresponds on the sky to a small circle (ring) of constant angular distance from the anti-CoE direction. Each ring is drawn with a band of half-width $\pm 9^\circ$ (18° end to end, i.e. 5% of 360°). The CoE axis is tilted by $+15^\circ$ out of the Galactic plane: the anti-CoE direction is at $(\ell, b) = (0^\circ, +15^\circ)$ and the CoE at $(180^\circ, -15^\circ)$. Sky maps use the Galactic Mollweide projection with longitude increasing to the left, as in Migkas et al. Fig. 8.

4 Results

4.1 Longitude profile

Figure 5 shows the model profile obtained by integrating the enclosed-mass acceleration along the flow for $d/D = 0.1$, compared with a single impulsive approximation and with the digitized data. The model reproduces the two-maxima, two-minima structure of the data, but with extrema at $\phi \simeq 0^\circ, 180^\circ$ and $90^\circ, 270^\circ$ fixed by axisymmetry. The axisymmetric model profile attains $R^2 = 0.307$ (rms residual 0.059), and a smooth sine-rotation of the same profile raises this to $R^2 = 0.553$ (rms 0.047; Fig. 6). An alternative evaluation of the first-principles geometry without any rotation template gives $R^2 = 0.543$.

The phase-rotation template of Eq. (3.3) attains its best agreement at $\mathrm{Amp} = -0.06$, $\varphi \simeq 11.5^\circ$, with $R^2 = 0.746$ (rms 0.038). The dependence of R^2_A on the template parameters is shown in Fig. 7(a): the agreement forms a broad ridge at $\mathrm{Amp} \approx -0.04$ to -0.08 and $\varphi \lesssim 30^\circ$, and degrades steadily toward larger phases. Applying the same template to the slope B of the L_X–T relation with an independent amplitude Amp_B [Fig. 7(b)] gives a best fit of $R^2_B = 0.790$ at $\mathrm{Amp}_B = -0.06$, $\varphi \simeq 185^\circ$; the nearly equivalent solution $\mathrm{Amp}_B = +0.06$ at small phase ($R^2_B = 0.790$) corresponds to $\mathrm{Amp}_B = -\mathrm{Amp}_A$ at the phase preferred by A. Adding the second-harmonic correction of Eq. (3.4) with $\varepsilon = -0.043$ and $\psi = 175.8^\circ$ yields $R^2 = 0.875$ and an rms residual of 0.029 (Fig. 8). The corrected profile has maxima at $\theta \simeq 30^\circ$ and 185° and minima at $\theta \simeq 115^\circ$ and 265°.

To isolate the contribution of the geometry, Fig. 9 compares the fully corrected profile with the product of the two correction factors applied to a constant, $A = 1 \times F_{\rm rot} \times F_{2\ell}$, where $F_{\rm rot}$ is the ratio of the rotated to the unrotated axisymmetric profile and $F_{2\ell}$ is the factor of Eq. (3.4). The squared correlation with the data is 0.366 for the correction factors alone and 0.875 for the full model. The correction factors on their own therefore do not account for the observed profile; the axisymmetric Sedov–Taylor/Gauss geometry carries most of the explained variance.

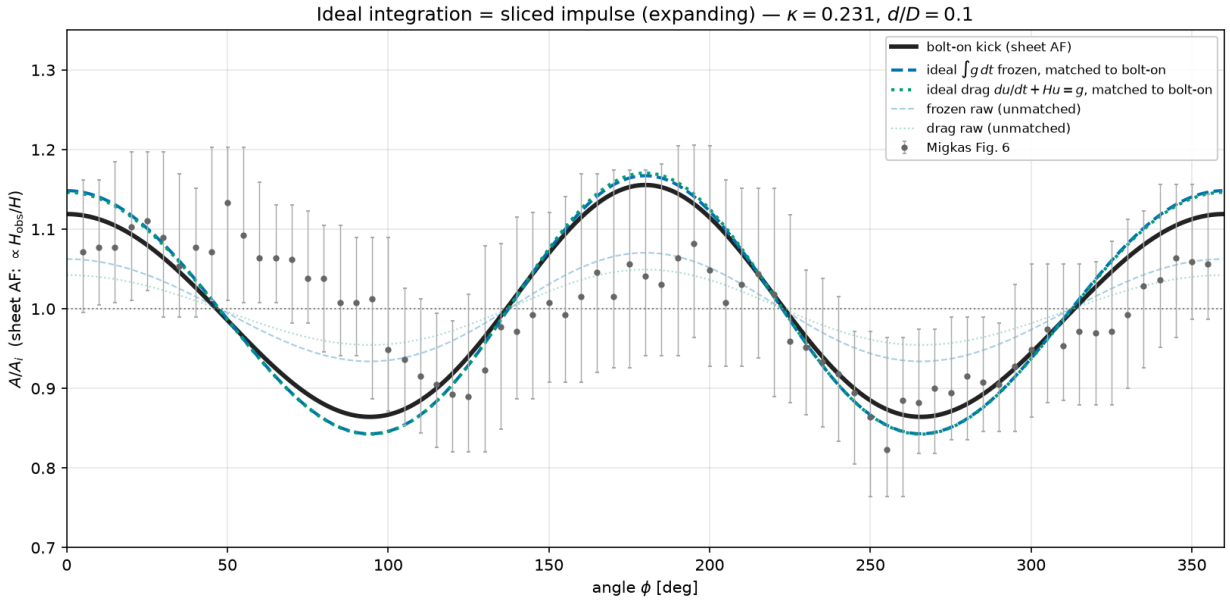


Figure 5: Model $A/A_{\rm all} \propto H_{\rm obs}/H$ versus sky angle ϕ for $d/D = 0.1$ and $\kappa = 0.231$. Solid: single impulsive deceleration. Dashed and dotted: time-integrated enclosed-mass acceleration (frozen and drag-coupled integration, respectively), normalized to the same amplitude; light curves show the unnormalized integrations. Points: digitized Migkas et al. profile.

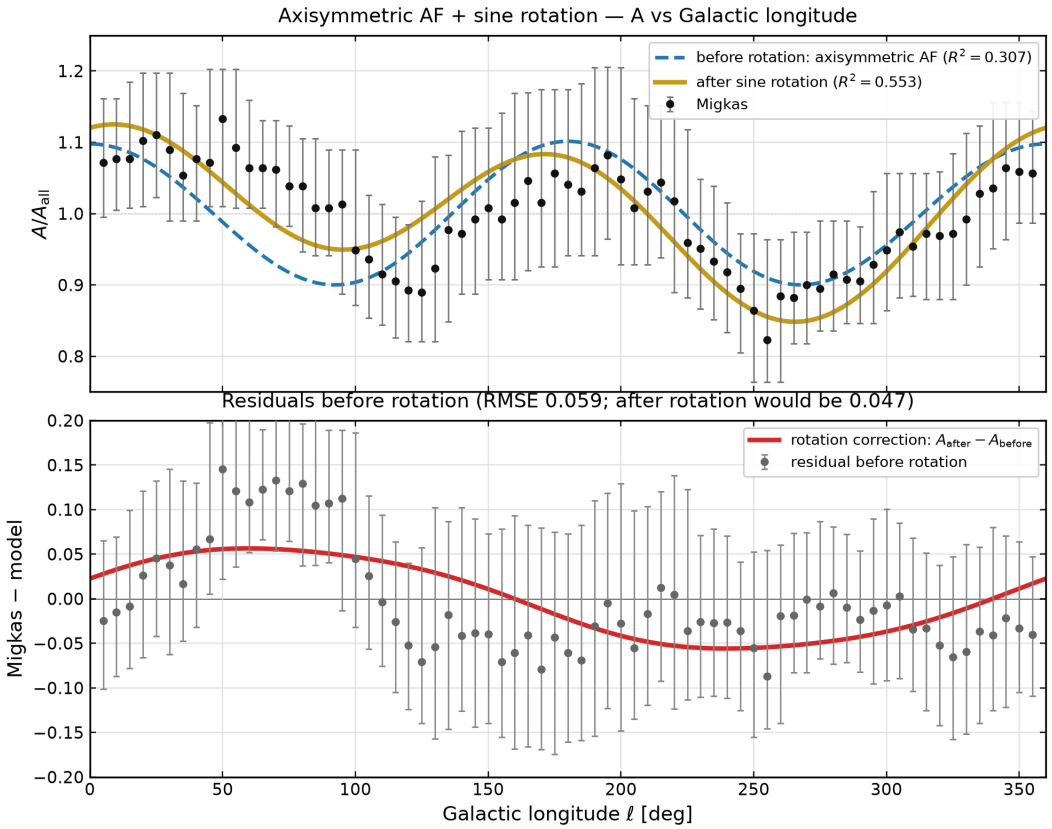
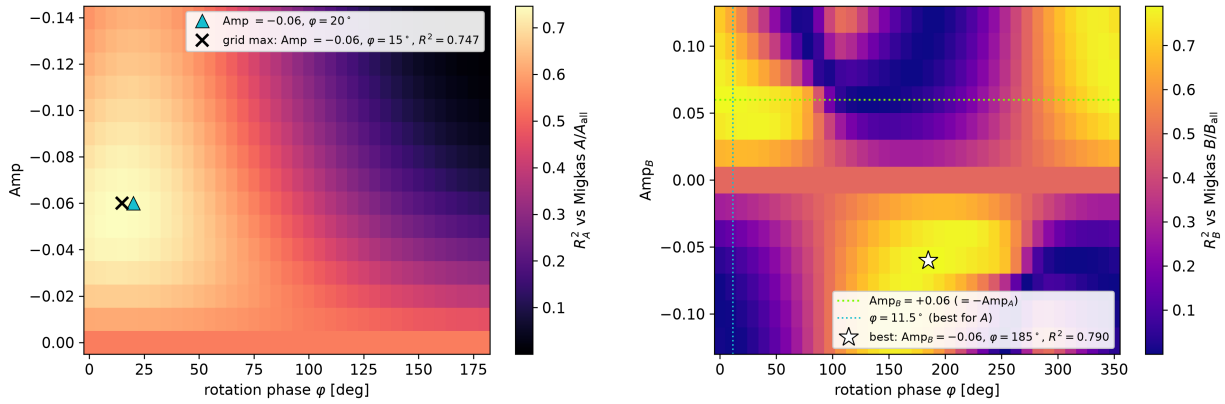


Figure 6: Axisymmetric model profile versus Galactic longitude before (dashed, $R^2 = 0.307$) and after (solid, $R^2 = 0.553$) a smooth sine rotation, compared with the digitized Migkas et al. profile. Lower panel: residuals before rotation (rms 0.059) and the rotation correction $A_{\rm after} - A_{\rm before}$.



(a) R_A^2 versus Amp and φ.

(b) R_B^2 versus Amp_B and φ.

Figure 7: Coefficient of determination of the phase-rotation template, Eq. (3.3), as a function of amplitude and phase. (a) R_A^2 against the Migkas et al. normalization profile A/A_{all} on a grid of Amp $\in [-0.14, 0]$ (step 0.01; axis inverted) and $\varphi \in [0°, 180°]$ (step 5°). The triangle marks Amp $= -0.06$, $\varphi = 20°$; the cross marks the grid maximum, $R_A^2 = 0.747$ at Amp $= -0.06$, $\varphi = 15°$. The best value from the coarser parameter scan, $R_A^2 = 0.746$ at $\varphi = 11.5°$, lies on the same ridge. (b) R_B^2 against the slope profile B/B_{all} for an independent amplitude $\mathrm{Amp}_B \in [-0.12, 0.12]$ and $\varphi \in [0°, 350°]$. The star marks the best fit, $R_B^2 = 0.790$ at $\mathrm{Amp}_B = -0.06$, $\varphi = 185°$; the dotted horizontal line marks $\mathrm{Amp}_B = +0.06 = -\mathrm{Amp}_A$ and the dotted vertical line the phase $\varphi = 11.5°$ preferred by A.

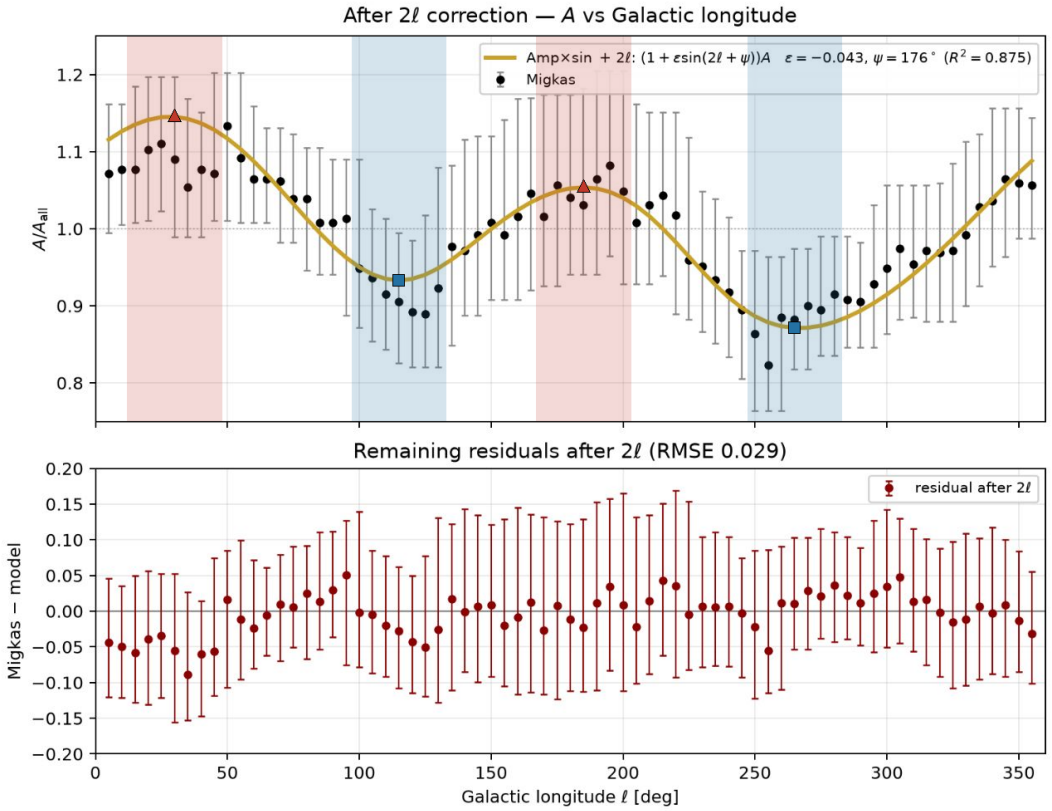


Figure 8: Model profile after the phase-rotation template and the second-harmonic correction (Eqs. 3.3–3.4; $\varepsilon = -0.043$, $\psi = 176°$), with $R^2 = 0.875$. Triangles and squares mark the maxima ($\theta \simeq 30°, 185°$) and minima ($\theta \simeq 115°, 265°$); shaded bands of $\pm 18°$ indicate their neighborhoods on the strip. Lower panel: residuals (rms 0.029).

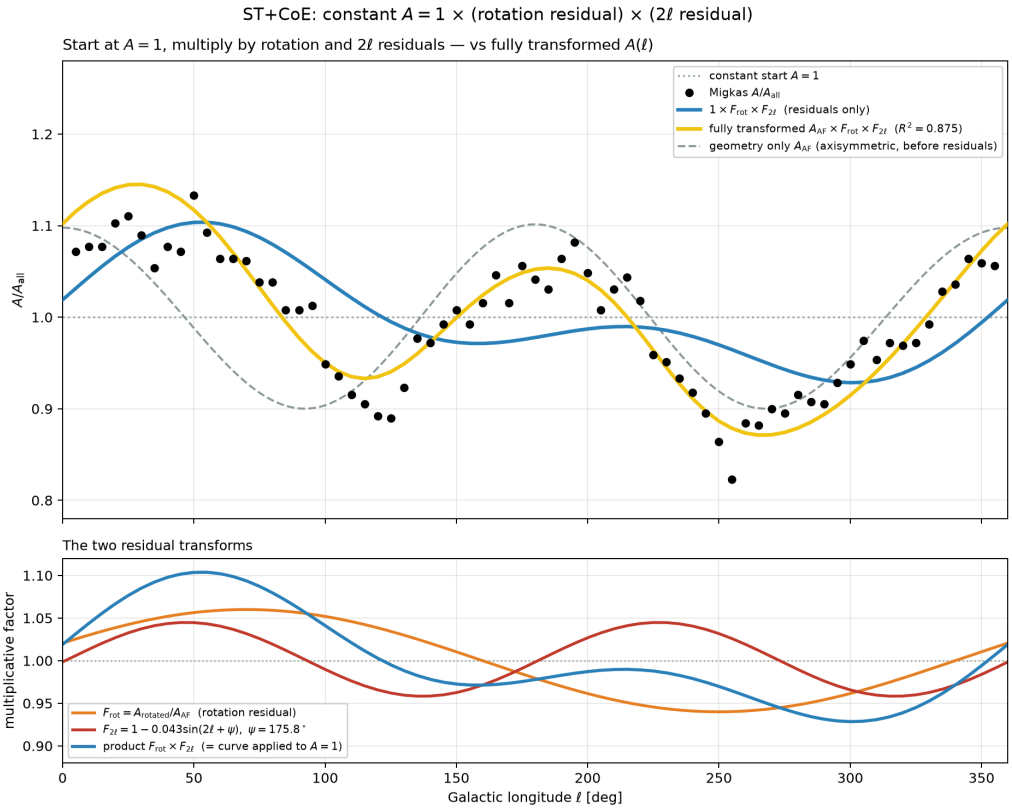


Figure 9: Decomposition of the corrected model. Upper panel: digitized data (points), axisymmetric geometry alone (dashed), the correction factors applied to a constant, $1 \times F_{\rm rot} \times F_{2\ell}$ (blue; squared correlation 0.366), and the full model $A_{\rm AF} \times F_{\rm rot} \times F_{2\ell}$ (yellow; $R^2 = 0.875$). Lower panel: the individual correction factors and their product.

4.2 Sky geometry of the extrema

Figure 10 shows the four extremal angles mapped to rings about the tilted CoE axis. Because the ring at angle θ and that at $360° - \theta$ coincide on the sky, the side of the axis on which the θ ray lies is indicated by drawing only the half of each ring on that side (the half satisfying $(\hat{n} \cdot \hat{Y}) \sin\theta \geq 0$, where $\hat{Y}$ is the in-plane direction perpendicular to the CoE axis). The angular radii of the rings about the anti-CoE direction are 30° (maximum), 175° (maximum, equivalently 5° about the CoE), 115° (minimum), and 95° (minimum).

Figure 11 overlays the same half rings on the Migkas et al. $\theta = 75°$ map of $A/A_{\rm all}$. The highest-normalization direction $(17°, -9°)$ lies 29.3° from the anti-CoE direction, within 1° of the $\theta = 30°$ maximum ring, and the lowest-normalization direction $(274°, -22°)$ lies 92.0° from the anti-CoE direction, within 3° of the $\theta = 265°$ minimum ring (angular radius 95°); both fall inside the ±9° bands. The secondary extrema of the longitude profile, marked on the Galactic plane, behave differently: $(115°, 0°)$ lies 114.1° from the anti-CoE direction, on the $\theta = 115°$ minimum ring, whereas $(185°, 0°)$ lies 164.2° from it, 11° from the center line of the $\theta = 185°$ maximum ring and just outside its band.

Figures 12 and 13 show the same geometry with full rings. The opacity of each ring decreases linearly with azimuth ψ around the ring, measured from the point on the θ-ray side, $\alpha \propto 1 - 0.88\,|\psi|/\pi$, so that each ring is darkest where the θ ray meets it and falls to 12% opacity at the diametrically opposite point. This representation shows the complete small circles while retaining the side information of the half-ring figures.

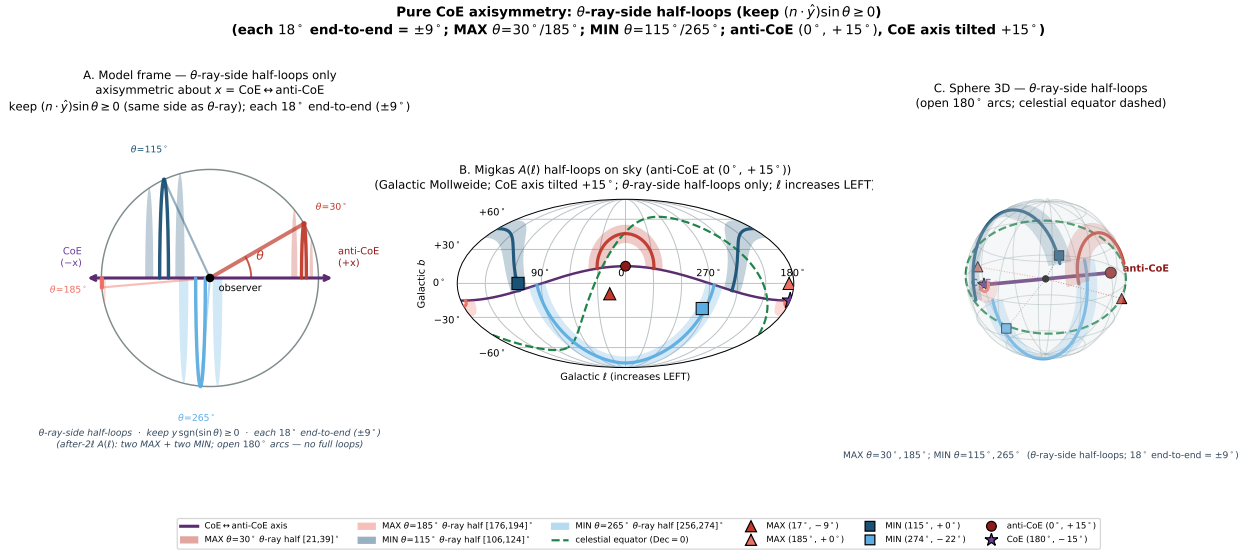


Figure 10: Sky geometry of the four extrema of the corrected longitude profile, with the CoE axis tilted by +15° (anti-CoE at $(\ell, b) = (0°, +15°)$, CoE at $(180°, -15°)$). Maxima at $\theta = 30°$ and 185° (red) and minima at $\theta = 115°$ and 265° (blue), measured from the anti-CoE direction, each with a band of ±9°. Only the half of each ring on the side of its θ ray is drawn. (A) Model frame, axisymmetric about the CoE–anti-CoE axis. (B) Galactic Mollweide projection, longitude increasing to the left; the dashed curve is the celestial equator. (C) Three-dimensional view of the celestial sphere. Triangles and squares mark the reference maxima at $(17°, -9°)$, $(185°, 0°)$ and minima at $(115°, 0°)$, $(274°, -22°)$.

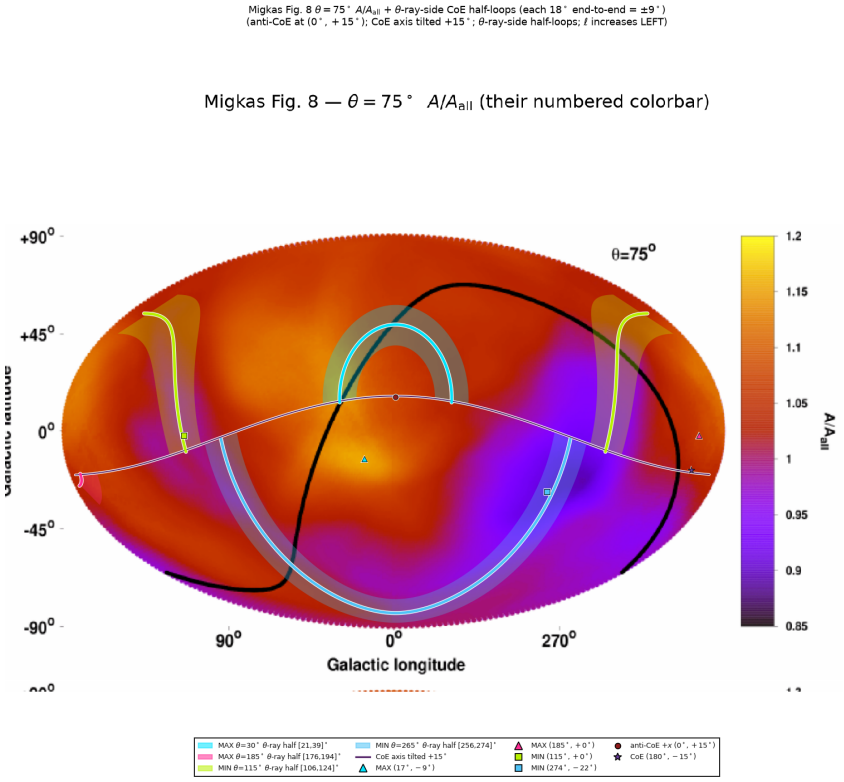


Figure 11: Half rings of Fig. 10 overlaid on the $\theta = 75°$ map of $A/A_{\rm all}$ from Migkas et al. [7] (their Fig. 8, reproduced with its original color scale). CoE axis tilted by $+15°$ (white-outlined line). Markers: maxima at $(17°, -9°)$ and $(185°, 0°)$; minima at $(115°, 0°)$ and $(274°, -22°)$.

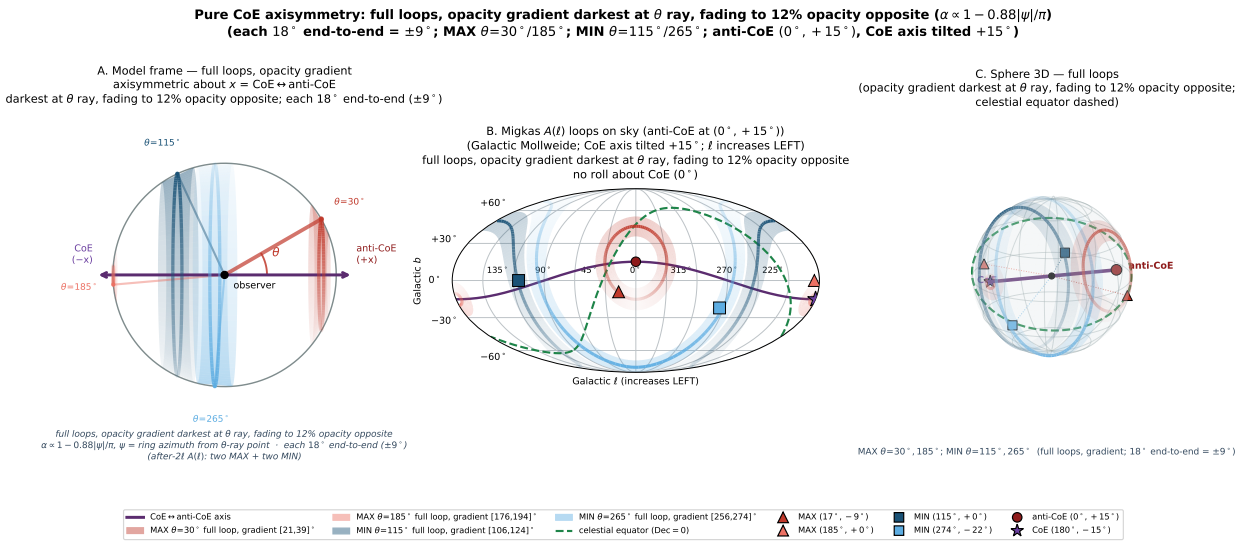


Figure 12: As Fig. 10, but showing the full rings with an opacity gradient: each ring is darkest at the point on the side of its θ ray and fades linearly in azimuth to 12% opacity at the opposite point ($\alpha \propto 1 - 0.88\,|\psi|/\pi$). Anti-CoE at $(0°, +15°)$, CoE at $(180°, -15°)$; maxima at $\theta = 30°, 185°$ and minima at $\theta = 115°, 265°$, each $\pm 9°$ wide. Panels (A)–(C) as in Fig. 10.

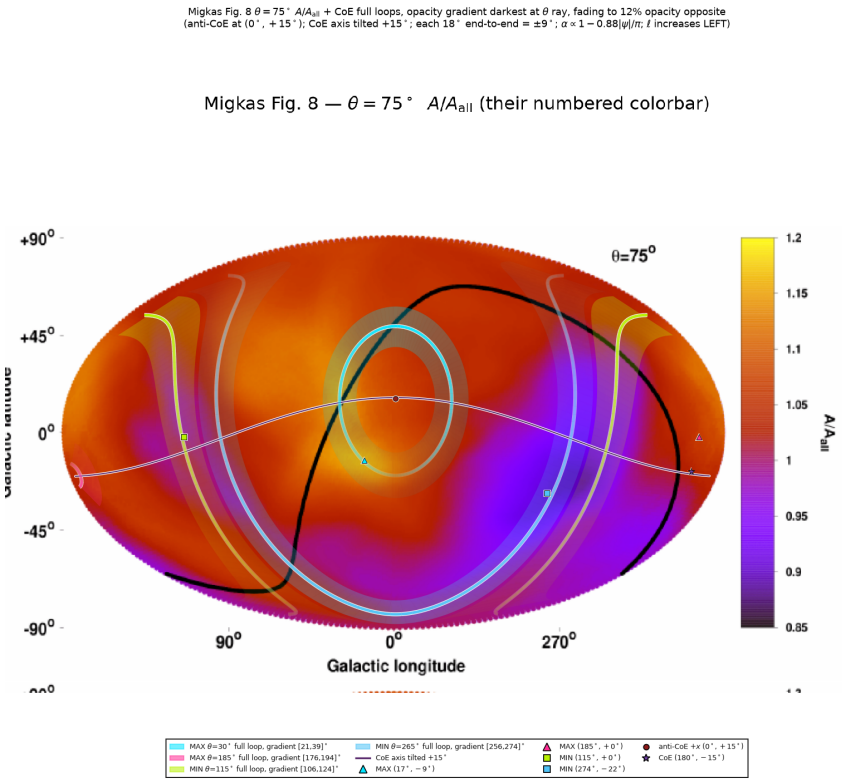
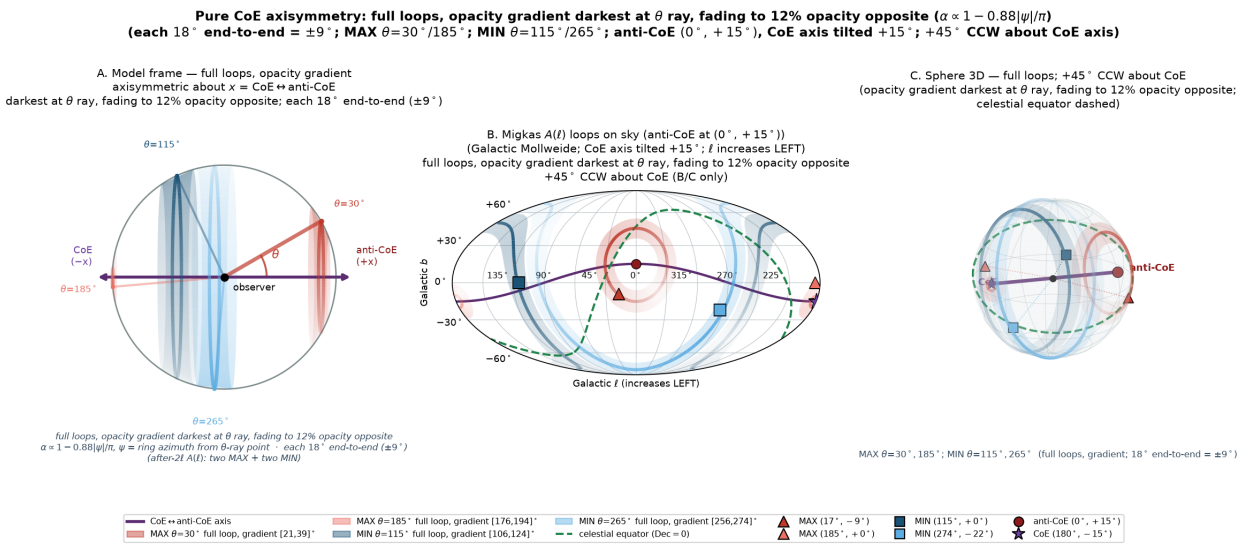
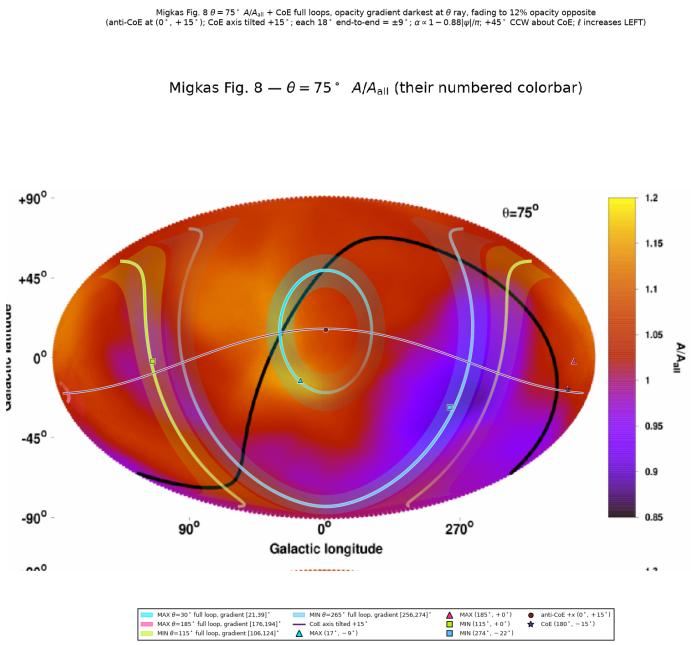


Figure 13: Full rings of Fig. 12, with the same opacity gradient, overlaid on the Migkas et al. [7] $\theta = 75°$ map of $A/A_{\rm all}$. Markers as in Fig. 11.

The orientation of the rings about the CoE axis is not fixed by the axisymmetric model in the model frame; only the axis direction is. As an illustration of the effect of this freedom on the sky projection, Fig. 14 shows the full-ring geometry with the sky panels rolled by 45° counter-clockwise about the CoE axis, the model-frame panel being left unchanged.



(a) Three-panel geometry; panels B and C rolled by 45° counter-clockwise about the CoE axis, panel A unrolled.



(b) Overlay on the Migkas et al. $\theta = 75°$ map with the same roll.

Figure 14: Full-ring geometry of Figs. 12 and 13 with the sky projection rolled by 45° counter-clockwise about the CoE axis (anti-CoE at $(0°, +15°)$, CoE at $(180°, -15°)$).

5 Discussion

Origin of the anisotropy. A central result of the model is that the displacement of the observer from the CoE is, by itself, unobservable in relative velocities: homologous expansion yields the same H in every direction for every observer (Eq. 2.7). Direction dependence is produced by the enclosed-mass gravity of a decreasing density profile, and at leading order it takes the form of a quadrupole about the CoE axis (Eq. 2.13). The empirical finding that the second harmonic dominates the Migkas et al. longitude profile (Eq. 3.1) is consistent with this structure. A pure dipole description, by contrast, accounts for only about a quarter of the variance.

Status of the templates. The phase-rotation template (Eq. 3.3) and the second-harmonic correction (Eq. 3.4) are introduced empirically and are not yet derived from the time-integrated enclosed-mass acceleration. Their parameters are fitted to the same data used to evaluate the agreement, so the improvement from $R^2 = 0.307$ to $R^2 = 0.875$ should not be regarded as a test of the model. The decomposition of Fig. 9 indicates that the correction factors alone account for a minority of the variance, but a first-principles derivation of any departure from axisymmetry—for example from a non-spherical mass distribution or from the time dependence of the flow—is required before the corrected profile can be regarded as a prediction.

Sky geometry. The ring construction translates the longitude profile into a two-dimensional prediction that can be compared with the Migkas et al. sky map. With the CoE axis tilted by 15°, the principal maximum and minimum rings pass close to the published extrema, while the ring associated with the secondary maximum at $(185°, 0°)$ is offset by about 11°. The comparison is at present qualitative; a quantitative test requires evaluating the model on the full sky with the survey footprint and cone-averaging used by Migkas et al., and assessing the significance of the alignment against randomly oriented axes.

Relation to the cosmic microwave background. The present work addresses the late-universe anisotropy of the Hubble flow as traced by the X-ray cluster scaling relation. A quantitative comparison with the cosmic microwave background, including its dipole and low multipoles, within the full model is deferred to future work. No conflict with CMB data is asserted or demonstrated here, and consistency is likewise not claimed.

Relativistic generalization. The dynamics have been treated in the Newtonian Sedov–Taylor limit with homologous interior flow. The relativistic blast-wave solutions of Blandford and McKee [3], the cosmological self-similar shocks of Bertschinger [4], and the general-relativistic shock-wave cosmologies of Smoller and Temple [5, 6] provide the frameworks within which the model can be extended to relativistic velocities and to distances comparable to or beyond the Hubble length, where the CoE is assumed to lie.

6 Conclusions

We have described a model in which the cosmic expansion is the interior flow of a Sedov–Taylor self-similar blast wave, $R \propto t^{2/(5-w)}$ for an ambient density $\rho \propto r^{-w}$, launched from a Center of Expansion beyond the horizon. Our conclusions are as follows.

1. Homologous expansion gives an isotropic apparent Hubble law for any observer, independent of the observer's distance from the CoE.

2. Enclosed-mass gravity, $g = -GM(< r)/r^2$, in a centrally concentrated, outward-decreasing density profile breaks homology and produces a direction-dependent $H_{\rm obs}$ whose leading term is a quadrupole about the CoE axis, with natural strength $\kappa = \Omega_m/2 = 0.15$.

3. The Migkas et al. $A(\ell)$ profile is dominated by its second harmonic: a dipole-plus-quadrupole fit gives $R^2 = 0.814$ against 0.271 for a dipole alone.

4. The axisymmetric model gives $R^2 = 0.307$; with empirical phase-rotation and second-harmonic corrections the agreement reaches $R^2 = 0.875$.

5. Mapped onto the sky about a CoE axis tilted by 15°, with the anti-CoE direction at $(0°, +15°)$, the principal extremal rings pass within a few degrees of the published maximum and minimum of the A map.

The next steps are a first-principles derivation of the corrections from time-integrated gravity, a full-sky comparison including the survey footprint, and a quantitative treatment of the cosmic microwave background within the model.

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The author thanks Wolfgang Rindler and Luke Ackerman for the early discussions in which the Sedov–Taylor centered-universe idea was developed. The sky maps in Figs. 11, 13 and 14 use the $\theta = 75°$ normalization map published by Migkas et al. [7].

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